

PROOF OF EXAMPLE 9.2.4 (3) (AN OPEN IMMERSION IS ÉTALE)

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The purpose of this note is to prove that if $f : R \rightarrow S$ is a homomorphism of commutative rings and $f^\# : \operatorname{Spec} S \rightarrow \operatorname{Spec} R$ is an open immersion, then S is an R -algebra of finite presentation (see Corollary 10 below). This theorem is referenced without proof in [1, Example 9.2.4 (3)]. The corollary and its proof presented here are based on [3, Proposition I.4.6, p. 328]. The proof below utilizes only theorems proven in [2] and [1].

Throughout, all rings are commutative. Let M be an R -module and α a nonzero element of R . Denote by M_α the localization of M with respect to the multiplicative set $\{1, \alpha, \alpha^2, \alpha^3, \dots\}$.

Definition 1. Let R be a commutative ring and M an R -module. We say M is *locally finitely generated*, if there exist elements $\alpha_1, \dots, \alpha_n$ in R such that $R = R\alpha_1 + \dots + R\alpha_n$ and for each i , M_{α_i} is a finitely generated R_{α_i} -module.

Lemma 2. *Let M be an R -module. If M is locally finitely generated, then M is finitely generated.*

Proof. Let $\alpha_1, \dots, \alpha_n$ in R such that $R = R\alpha_1 + \dots + R\alpha_n$ and for each i , M_{α_i} is a finitely generated R_{α_i} -module. For each i , let $\{x_{ij} \mid 1 \leq j \leq n_i\}$ be a subset of M which is a generating set for M_{α_i} as an R_{α_i} -module. The reader should verify that $\bigcup_i \{x_{ij} \mid 1 \leq j \leq n_i\}$ is a generating set for the R -module M . $\square$

Definition 3. Let S be a commutative R -algebra with structure homomorphism $\theta : R \rightarrow S$. We say S is an *R -algebra of finite presentation*, if there is a polynomial ring in $n \geq 1$ variables over R and a surjective R -algebra homomorphism $\phi : R[x_1, \dots, x_n] \rightarrow S$ such that the kernel of ϕ is a finitely generated ideal. The R -algebra S is said to be *locally of finite presentation*, if there exist elements $\beta_1, \dots, \beta_n$ in S such that $S = S\beta_1 + \dots + S\beta_n$ and for each i , S_{β_i} is an R -algebra of finite presentation. We also say θ is *locally of finite presentation*.

Example 4. Let α be a non-zero element of R . Define $\phi : R[x] \rightarrow R_\alpha$ by $\phi(x) = 1/\alpha$. The kernel of ϕ is the principal ideal generated by $x\alpha - 1$. This shows R_α is an R -algebra of finite presentation.

Lemma 5. (1) (*Change of Base*) Let $f : R \rightarrow T$ be a homomorphism of commutative rings such that T is an R -algebra of finite presentation. If $g : R \rightarrow S$ is any homomorphism of commutative rings, then $f \otimes 1 : S \rightarrow T \otimes_R S$ makes $T \otimes_R S$ into an S -algebra of finite presentation.
 (2) (*Finitely Presented over Finitely Presented is Finitely Presented*) If S is a finitely presented R -algebra and T is a finitely presented S -algebra, then T is a finitely presented R -algebra.

Proof. (1): There exists an onto homomorphism $\phi : R[x_1, \dots, x_n] \rightarrow T$ such that $\ker \phi$ is a finitely generated ideal. Then $\phi \otimes 1 : S[x_1, \dots, x_n] \rightarrow T \otimes_R S$ is onto. The kernel of $\phi \otimes 1$ is generated by the image of $\ker \phi \otimes_R S$, hence is a finitely generated ideal.

(2): There exist homomorphisms $\phi : R[x_1, \dots, x_m] \rightarrow S$ and $\psi : S[y_1, \dots, y_n] \rightarrow T$ such that ϕ and ψ are both onto, and $\ker \phi$ and $\ker \psi$ are both finitely generated ideals. Define $\chi : R[x_1, \dots, x_m][y_1, \dots, y_n] \rightarrow T$ by $\chi(x_i) = \psi\phi(x_i)$ for each i and $\chi(y_j) = \psi(y_j)$ for each j . Then χ is onto and the kernel of χ is generated by $\ker \phi + \ker \psi$, hence is finitely generated. $\square$

Proposition 6. *Let $f : R \rightarrow S$ be a homomorphism of commutative rings. Prove the following.*

- (1) *If the kernel of f is a finitely generated ideal and f is onto, then S is an R -algebra of finite presentation.*
- (2) *If f is one-to-one and S is a finitely generated R -module, then S is an R -algebra of finite presentation.*
- (3) *If the kernel of f is a finitely generated ideal and S is a finitely generated R -module, then S is an R -algebra of finite presentation.*

Proof. The proof is left to the reader as an exercise. $\square$

Proposition 7. (1) *If $f : R \rightarrow S$ is a local isomorphism, then f is locally of finite presentation.*

- (2) *(Composition) If $f : R \rightarrow S$ and $g : S \rightarrow T$ are both locally of finite presentation, then $g \circ f : R \rightarrow T$ is locally of finite presentation.*
- (3) *(Change of Base) Let $f : R \rightarrow T$ and $g : R \rightarrow S$. If f is locally of finite presentation, then $f \otimes 1 : S \rightarrow T \otimes_R S$ makes $T \otimes_R S$ into an S -algebra that is locally of finite presentation.*
- (4) *(Tensor Product) Let $f : S \rightarrow S'$ and $g : T \rightarrow T'$ be R -algebra homomorphisms. If f and g are both locally of finite presentation, then $f \otimes g : S \otimes_R T \rightarrow S' \otimes_R T'$ is locally of finite presentation.*
- (5) *(Descent) If $f : R \rightarrow S$ is locally of finite type and $g : S \rightarrow T$, and $g \circ f : R \rightarrow T$ is locally of finite presentation, then g is locally of finite presentation.*

Proof. (1): For every $\mathfrak{q} \in \operatorname{Spec} S$ there exists $\alpha \in R$ such that $f(\alpha) \in S - \mathfrak{q}$ and $R_\alpha \rightarrow S_\alpha$ is an isomorphism. By Example 4, S_α is an R -algebra of finite presentation. By [2, Exercise 3.3.29], S is locally of finite presentation.

(2): Let $\mathfrak{r} \in \operatorname{Spec} T$ be arbitrary. There exists $\beta \in T - \mathfrak{r}$ such that $g : S \rightarrow T_\beta$ is of finite presentation. Let $\mathfrak{q} = \mathfrak{r} \cap S$. There exists $\alpha \in S - \mathfrak{q}$ such that $f : R \rightarrow S_\alpha$ is of finite presentation. By Lemma 5(1), $g : S_\alpha \rightarrow T_{\beta\alpha}$ is of finite presentation. By Lemma 5(2), $g \circ f : R \rightarrow S_\alpha \rightarrow T_{\beta\alpha}$ is of finite presentation. Because $\operatorname{Spec} S$ is compact ([2, Exercise 3.3.29]), we see that T is locally of finite presentation as an R -algebra.

(3): Start with elements $\beta_1, \dots, \beta_n$ in T such that $T = T\beta_1 + \dots + T\beta_n$ and for each i , T_{β_i} is an R -algebra of finite presentation. Then $\beta_1 \otimes 1, \dots, \beta_n \otimes 1$ generate the unit ideal in $T \otimes_R S$. By Lemma 5(1), $T_{\beta_i} \otimes_R S$ is an S -algebra of finite presentation, for each i .

(4): By (3), $f \otimes 1 : S \otimes_R T \rightarrow S' \otimes_R T$ and $1 \otimes g : S' \otimes_R T \rightarrow S' \otimes_R T'$ are both locally of finite presentation. By (2), $f \otimes g$ is locally of finite presentation.

(5): First note that $S \otimes_{S \otimes_R S} (T \otimes_R S)$ is isomorphic to T . The diagram

$$\begin{array}{ccc}
 & T & \\
 \rho \nearrow & & \nwarrow g \\
 T \otimes_R S & & S \\
 g \otimes 1 \nwarrow & & \nearrow \mu \\
 & S \otimes_R S &
 \end{array}$$

commutes, where $\rho(t \otimes s) = ts$. By [2, Proposition 3.5.14], f makes S into a finitely generated R -algebra. By [2, Lemma 10.1.6], the multiplication homomorphism $\mu : S \otimes_R S \rightarrow S$ makes S into a finitely presented $S \otimes_R S$ -module. By Proposition 6, S is a finitely presented $S \otimes_R S$ -algebra. By (3), ρ makes T into a $T \otimes_R S$ -algebra of finite presentation. By (3) again, $(g \circ f) \otimes 1 : S \rightarrow T \otimes_R S$ is locally of finite presentation. The diagram

$$\begin{array}{ccc}
 S & \xrightarrow{g} & T \\
 (g \circ f) \otimes 1 \searrow & & \nearrow \rho \\
 & T \otimes_R S &
 \end{array}$$

commutes, where $\rho(t \otimes s) = ts$. By (2), g is locally of finite presentation. $\square$

Lemma 8. *Let I be a proper ideal in R and α a nonzero element of R such that R_α/I_α is an R -algebra of finite presentation. Then I_α is a finitely generated ideal in R_α .*

Proof. There exists an onto R -algebra homomorphism $\phi : R[x_1, \dots, x_n] \rightarrow R_\alpha/I_\alpha$ such that $\ker \phi$ is a finitely generated ideal. Tensor with $(\cdot) \otimes_R R_\alpha$ to get the exact sequence

$$0 \rightarrow \ker \phi \otimes_R R_\alpha \rightarrow R_\alpha[x_1, \dots, x_n] \xrightarrow{\tau} R_\alpha/I_\alpha \rightarrow 0$$

where $\tau = \phi \otimes 1$. Therefore, $\ker \tau = \ker \phi \otimes_R R_\alpha$ is a finitely generated ideal. Let $\eta : R_\alpha \rightarrow R_\alpha/I_\alpha$ be the natural map. Since τ is an R_α -algebra homomorphism, there exists a lifting $\psi : R_\alpha[x_1, \dots, x_n] \rightarrow R_\alpha$ such that $\eta\psi(x_i) = \tau(x_i)$ for each i . The diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \ker \tau & \longrightarrow & R[x_1, \dots, x_n] & \xrightarrow{\tau} & R_\alpha/I_\alpha \longrightarrow 0 \\
 & & \downarrow & & \downarrow \psi & & \downarrow u \\
 0 & \longrightarrow & I_\alpha & \longrightarrow & R_\alpha & \xrightarrow{\eta} & R_\alpha/I_\alpha \longrightarrow 0
 \end{array}$$

commutes and the rows are exact sequences. Since ψ is onto and u is the identity map, the Snake Lemma ([2, Theorem 2.3.2]) implies $\psi(\ker \tau) = I_\alpha$. This proves I_α is a finitely generated ideal in R_α . $\square$

Proposition 9. *If $f : R \rightarrow S$ is locally of finite presentation, then f is of finite presentation.*

Proof. There is an onto R -algebra homomorphism $\theta : R[x_1, \dots, x_n] \rightarrow S$. Therefore f factors through θ and the diagram

$$\begin{array}{ccc} R[x_1, \dots, x_n] & \xrightarrow{\theta} & S \\ \sigma \uparrow & \nearrow f & \\ R & & \end{array}$$

commutes where σ is the natural map. Let J denote the kernel of θ . By Proposition 7 (5), θ is locally of finite presentation. By Lemma 8, J is locally a finitely generated ideal in $R[x_1, \dots, x_n]$. By Lemma 2, J is a finitely generated ideal. Hence, S is an R -algebra of finite presentation. $\square$

Corollary 10. *Let $f : R \rightarrow S$ be a homomorphism of commutative rings. If $f^\# : \operatorname{Spec} S \rightarrow \operatorname{Spec} R$ is an open immersion, then S is an R -algebra of finite presentation.*

Proof. By Proposition 7 (1), $f : R \rightarrow S$ is locally of finite presentation. By Proposition 9, $f : R \rightarrow S$ is of finite presentation. $\square$

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